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Master Thesis

2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach

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A thesis submitted in fulfillment of the requirements for the degree of Master of Science

February 19, 2026

Contents

Notation and conventions

Throughout this thesis, natural units are used such that $\hbar = 1$. Frequencies, energies, and inverse times are therefore expressed in the same units.

The following notation conventions apply:

- **Vectors.** Vectors in real space are denoted by bold symbols, e.g. $\boldsymbol{E}$ (electric field), $\boldsymbol{P}$ (polarization), $\boldsymbol{k}$ (wave vector), $\boldsymbol{d}$ (transition dipole moment), and $\boldsymbol{e}$ (polarization unit vector). The magnitude of a vector $\boldsymbol{d}$ is written as $d = |\boldsymbol{d}|$ when required.

- **Operators and states.** Operators acting on Hilbert spaces carry a hat, e.g. $\hat{H}$ for the Hamiltonian and $\hat{\boldsymbol{\mu}}$ for the dipole operator. Density operators are written as $\hat{\rho}$. Matrix elements in a chosen basis are denoted by plain symbols, e.g. $\rho_{ij} = \langle i|\hat{\rho}|j\rangle$.

- **Quantum mechanical pictures.** Different dynamical representations (Schrödinger, interaction, Heisenberg) are indicated by subscripts. For example, $\hat{O}_I(t)$ denotes an operator in the interaction picture. To avoid ambiguity with this convention, the interaction Hamiltonian is written as $\hat{H}_{\text{int}}$.

- **Superoperators.** Linear maps acting on operators (e.g. propagators or Liouvillians) are written in calligraphic font, such as $\mathcal{U}$, $\mathcal{L}$, or $\mathcal{M}$. Their action on density operators is written explicitly as $\mathcal{L}[\hat{\rho}]$.

- **Field projections.** For a fixed laboratory polarization direction $\boldsymbol{e}$, vector fields are often projected onto this axis. The scalar field envelope is then defined as

  $$\varepsilon(t) = \boldsymbol{e} \cdot \boldsymbol{E}(t),$$

  and correspondingly $\hat{\mu} = \hat{\boldsymbol{\mu}} \cdot \boldsymbol{e}$. Whenever such projections are used, the underlying vector character is understood but suppressed in the notation.

- **Complex quantities.** The symbols Re and Im denote the real and imaginary parts of a complex quantity, respectively.

Chapter 1

Open Quantum Systems

Any real-world quantum system is not perfectly isolated. Instead, it is coupled to surrounding degrees of freedom such as electromagnetic modes, lattice vibrations, or solvent fluctuations, though the coupling is often weak but never exactly zero. In contrast to an ideal closed system with unitary Schrödinger evolution,

$$i\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle = \hat{H}(t)\left|\psi(t)\right\rangle, \tag{1.1}$$

an open system generally exhibits effective non-unitary dynamics once the environment is traced out. This coupling drives relaxation toward an environment-defined equilibrium and leads to two key mechanisms: **decoherence**, the loss of phase relations and dephasing, and **dissipation**, energy exchange and thermalization [1, 2].

Concrete examples include atoms subject to vacuum electromagnetic field fluctuations [1], quantum dots in solid-state settings coupled to phonons [3], molecular systems interacting with surrounding solvent degrees of freedom [4], and quantum computers whose coherence can be rapidly degraded by environmental noise [5].

For molecular chemistry and biological complexes, the relevant quantum degrees of freedom are embedded in warm, wet, and structurally complex environments, so strong decoherence and fast relaxation are expected. Yet these systems must still obey quantum laws, so the task is to explain classical-looking behavior within a fully quantum description. [6].

This chapter provides the theoretical foundation for including environmental effects in the quantum-dynamical models used throughout the thesis. A practical weak-coupling description is introduced that connects microscopic Hamiltonians to relaxation/dephasing rates and to bath models used later in applications and simulations. In Sec. 1.1 the Redfield master equation is introduced and derived from microscopic system–bath dynamics. In Sec. 1.2 bath correlation functions and their Fourier/spectral representations are discussed, including how emission and absorption arise via detailed balance. In Sec. 1.3 harmonic-oscillator (bosonic) environments are considered and explicit thermal correlators are computed, establishing the link to spectral densities. Finally, Sec. 1.4 summarizes common phenomenological spectral densities such as Ohmic-type and Drude–Lorentz and their physical interpretation.

1.1 The Redfield Master Equation

In this thesis the Redfield master equation is used as the weak-coupling description of open-system dynamics. Once the environment is specified by its correlation function or equivalently a spectral density, the reduced dynamics follows directly from the microscopic system–bath model and can be simulated efficiently even for multi-level molecular and excitonic systems [1, 2, 4]. More accurate methods such as numerically exact path-integral [7] or hierarchy techniques [8] can be too costly for the parameter scans and system sizes considered here. It is important to note that the Redfield equation, while preserving trace and Hermiticity, does not guarantee complete positivity of the density matrix, which is a fundamental requirement for physical quantum states [9]. So care must be taken when determining when the Redfield equation is useful [10–12].

Originally developed by A.G. Redfield in 1957 and 1965 for nuclear magnetic resonance relaxation phenomena [10], the Redfield equation describes the time evolution of the reduced density matrix, formally defined in Sec. 1.1.1, of a quantum system weakly coupled to a thermal environment. Unlike phenomenological approaches that introduce dissipation and dephasing *ad hoc*, the Redfield equation relates them to environmental correlation functions or spectral densities [1, 2, 13].

The following requirements must be fulfilled by the final derived Redfield equation:

1. The equation should be linear in the system density matrix $\frac{d}{dt}\hat{\rho}_S(t) = F(\hat{\rho}_S(t))$. This is the reduced equation of motion.
2. The equation should be Markovian, meaning that the evolution of the system density matrix at time t only depends on the state of the system at time t and not on its past history.
3. The equation should be trace-preserving. $\mathrm{Tr}[\hat{\rho}_S(t)] = \mathrm{Tr}[\hat{\rho}_S(0)]$ for all times t.

The Redfield equation is now derived from the microscopic dynamics of the system–environment composite, following the approach presented in [14] and standard textbooks such as Breuer and Petruccione [1]. The steps are presented in a more explicit way. In particular, the elimination of the first-order contribution is treated rigorously. The environmental correlation functions and spectral densities that characterize the bath properties and determine the system's relaxation behavior are then examined.

1.1.1 Mathematical Background

Before the derivation of the Redfield equation, the basic mathematical tools and notation used throughout are briefly collected: the density-operator formalism for mixed states, the trace and partial trace for extracting reduced system dynamics, and the corresponding expectation values.

Density Matrix Formalism

The density matrix formalism provides a powerful framework to describe the dynamics of quantum systems, especially when dealing with mixed states and decoherence effects. The density matrix $\hat{\rho}$ is defined as [15]

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|, \quad p_i \geq 0, \quad \sum_i p_i = 1. \tag{1.2}$$

It represents a statistical mixture, the average of an ensemble with different realizations of pure quantum states ψ_i with classical probabilities p_i. In other words, one cannot tell with certainty the quantum state of the system. This is particularly useful in the context of open quantum systems, where the system is entangled with an environment and one loses information about the system of interest to the total system. The conditions in Eq. (1.2) ensure the following properties that a physical density matrix must satisfy

$$\begin{aligned} \hat{\rho}^\dagger &= \hat{\rho}, \\ \left\langle\psi\right| \hat{\rho} \left|\psi\right\rangle &\geq 0 \quad \forall \left|\psi\right\rangle, \\ \mathrm{Tr}(\hat{\rho}) &= 1. \end{aligned}$$

In this formalism, for a canonical basis $\{|i\rangle\}$, the diagonal elements ρ_{ii} represent populations and the off-diagonal elements ρ_{ij} represent coherences between states. Coherences are phase relations, which are crucial for interference. For example for a two level system a clear phase relation and a pure quantum state would have off diagonal elements of $\rho_{ij} = 1/2$. This state is often referred to as a fully coherent superposition of the two states.

When coupling a system to an environment, the environment is responsible for decoherence. The state evolves over time to a purely statistical mixture of states.

The mean value or expectation value of an operator $\hat{O}$ can be calculated by additionally averaging over the classical probabilities p_i to obtain

$$\begin{aligned}\langle \hat{A} \rangle &= \sum_i p_i \langle \psi_i | \hat{A} | \psi_i \rangle \\ &= \sum_n \sum_i p_i \langle \psi_i | \hat{A} | \phi_n \rangle \langle \phi_n | \psi_i \rangle \\ &= \sum_n \langle \phi_n | \sum_i | \psi_i \rangle p_i \langle \psi_i | \hat{A} | \phi_n \rangle \\ &= \mathrm{Tr}(\hat{\rho} \hat{A}).\end{aligned}$$

Here, an arbitrary basis $\{|\phi_n\rangle\}$ and a corresponding completeness relation $\sum_n |\phi_n\rangle \langle \phi_n| = \mathbb{1}$ was used.

Partial Trace

The partial trace simplifies a quantum system description by removing certain parts through averaging. It can be viewed as the inverse operation to forming a tensor-product space. This is useful when only one part of a composite system is of interest. For a bipartite Hilbert space $\mathcal{H}_\mathcal{A} \otimes \mathcal{H}_\mathcal{B}$, the partial trace over $\mathcal{H}_\mathcal{B}$ is the unique linear map $\mathrm{Tr}_B : \mathcal{B}_1(\mathcal{H}_\mathcal{A} \otimes \mathcal{H}_\mathcal{B}) \to \mathcal{B}_1(\mathcal{H}_\mathcal{A})$ satisfying $\mathrm{Tr}[(\mathrm{Tr}_B \hat{X})\, \hat{A}] = \mathrm{Tr}[\hat{X}(\hat{A} \otimes \mathbb{1}_B)]$ for all $\hat{A} \in \mathcal{B}(\mathcal{H}_\mathcal{A})$, where $\mathcal{B}(\mathcal{H})$ is the algebra of bounded operators on $\mathcal{H}$, and $\mathcal{B}_1$ denotes trace-class operators, namely those X with finite trace norm[1, 16]

$$\|\hat{X}\|_1 = \mathrm{Tr}\sqrt{\hat{X}^\dagger \hat{X}}$$

This ensures expectation values of local observables $\hat{A} \otimes \mathbb{1}_B$ are preserved. In an orthonormal basis $\{|b_i\rangle\}$ of $\mathcal{H}_\mathcal{B}$, it acts as

$$\mathrm{Tr}_B \hat{X} = \sum_i (\mathbb{1}_A \otimes \langle b_i|)\, \hat{X}\, (\mathbb{1}_A \otimes |b_i\rangle). \tag{1.3}$$

With this definition thermal expectation values take the form

$$\langle \hat{A} \rangle = \mathrm{Tr}[\hat{\rho}\, \hat{A}] = \frac{1}{Z} \sum_n e^{-\beta E_n} A_{nn}, \tag{1.4}$$

with the inverse temperature

$$\beta = \frac{1}{k_\mathrm{B} T}.$$

1.1.2 System–Environment Setup

A quantum system of interest interacting with an environment with effectively infinitely many degrees of freedom is considered. The total Hilbert space $\mathcal{H}_T$ is the tensor product of the Hilbert space of the system $\mathcal{H}_S$ and that of the environment $\mathcal{H}_E$:

$$\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E.$$

The evolution of the total system is governed by the Liouville–von Neumann equation

$$\frac{d}{dt} \hat{\rho}_T(t) = -i[\hat{H}_T, \hat{\rho}_T(t)], \tag{1.5}$$

where $\hat{\rho}_T(t)$ is the density matrix of the total system and $\hat{H}_T$ is the total Hamiltonian. Units with $\hbar = 1$ are used throughout this thesis. Equation (1.5) is the mixed-state analogue of the Schrödinger equation Eq. (1.1) (see Sec. 1.1.1). Without loss of generality, the total Hamiltonian $\hat{H}_T \in \mathcal{B}(\mathcal{H}_T)$ can be decomposed as

$$\hat{H}_T = \hat{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \hat{H}_E + \alpha \hat{H}_{\text{int}},$$

where $\hat{H}_S$ and $\hat{H}_E$ act on the individual Hilbert spaces $\mathcal{H}_S$ and $\mathcal{H}_E$, respectively, while $\hat{H}_{\text{int}}$ acts on the composite space $\mathcal{H}_T$ and describes the system–environment interaction. The dimensionless parameter α is the coupling strength parameter.

The open-system setup is sketched in Fig. 1.1.

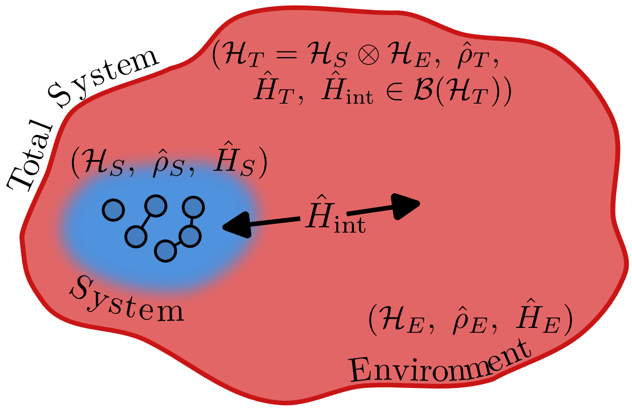

Figure 1.1: Schematic open-system setup. A system S is coupled to an environment E via $\hat{H}_{\text{int}}$. Schematically the system is indicated to contain six interacting particles such as atoms or molecules, while the environment has infinitely many degrees of freedom.

The interaction Hamiltonian is typically written in the form

$$\hat{H}_{\text{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i, \tag{1.6}$$

where $\hat{S}_i$ are system operators and $\hat{E}_i$ are environment operators. These will be concretized later in Sec. 1.3.

1.1.3 Interaction Picture

To describe the system dynamics, the interaction picture is used, where operators evolve with respect to the free Hamiltonian $\hat{H}_0 = \hat{H}_S + \hat{H}_E$. Any operator $\hat{O}$ in the Schrödinger picture takes the form

$$\hat{O}_I(t) = e^{i(\hat{H}_0)t}\, \hat{O}\, e^{-i(\hat{H}_0)t},$$

in the interaction picture. Throughout this thesis, hats indicate operators, while the subscript I indicates an interaction-picture quantity. States now evolve only according to the interaction Hamiltonian $\hat{H}_{\text{int}}$, and Eq. (1.5) becomes

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha[\hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(t)], \tag{1.7}$$

which can be formally integrated as

$$\hat{\rho}_{T,I}(t) = \hat{\rho}_{T,I}(0) - i\alpha \int_0^t ds [\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(s)].$$

Inserting this back into Eq. (1.7) yields

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha\left[\hat{H}_{\mathrm{int},I}(t), \hat{\rho}_{T,I}(0)\right] - \alpha^2\int_0^t \left[\hat{H}_{\mathrm{int},I}(t), \left[\hat{H}_{\mathrm{int},I}(s), \hat{\rho}_{T,I}(s)\right]\right] ds,$$

The iteration can be repeated, leading to a series expansion in powers of α

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha\left[\hat{H}_{\mathrm{int},I}(t), \hat{\rho}_{T,I}(0)\right] - \alpha^2\int_0^t \left[\hat{H}_{\mathrm{int},I}(t), \left[\hat{H}_{\mathrm{int},I}(s), \hat{\rho}_{T,I}(s)\right]\right] ds + \mathcal{O}(\alpha^3). \tag{1.8}$$

The expansion is truncated at second order, justified by the weak coupling assumption $\alpha \ll 1$, which represents the **Born approximation**.

Equation (1.8) still contains the full history through $\hat{\rho}_{T,I}(s)$ inside the integral and is therefore *non-Markovian*. As a first step toward a time-local Born–Markov description, one assumes that the total state varies slowly on the bath correlation time and replaces $\hat{\rho}_{T,I}(s) \to \hat{\rho}_{T,I}(t)$ inside the integrand. This is a time-local substitution at second order and yields

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha\big[\hat{H}_{\mathrm{int},I}(t), \hat{\rho}_{T,I}(0)\big] - \alpha^2\int_0^t ds\,\big[\hat{H}_{\mathrm{int},I}(t), [\hat{H}_{\mathrm{int},I}(s), \hat{\rho}_{T,I}(t)]\big], \tag{1.9}$$

which is now local in $\hat{\rho}_{T,I}(t)$ (but still retains an explicit upper integration limit t so is not yet Markovian). The remaining Markov step is the extension of the upper limit to ∞ using short bath memory.

The exact equation of motion for the reduced state $\hat{\rho}(t)$ involves the full many-body dynamics of the environment, which is generally intractable. Therefore, the next step is to trace out the environmental degrees of freedom.

1.1.4 Reduced Dynamics and Born Approximation

Since the dynamics of the system alone is of interest, the reduced density matrix is defined using the partial trace operation in Eq. (1.3):

$$\hat{\rho}_{S,I}(t) = \mathrm{Tr}_E[\hat{\rho}_{T,I}(t)].$$

Thus Eq. (1.9) becomes

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -i\,\alpha\,\mathrm{Tr}_E\big[\,\hat{H}_{\mathrm{int},I}(t), \hat{\rho}_{T,I}(0)\,\big] - \alpha^2\int_0^t ds\,\mathrm{Tr}_E\big[\,\hat{H}_{\mathrm{int},I}(t), [\hat{H}_{\mathrm{int},I}(s), \hat{\rho}_{T,I}(t)]\,\big]$$

Vanishing First-Order Term

The first order term in α in Eq. (1.10) will vanish after tracing over the environment if the interaction operators satisfy

$$\langle\hat{E}_i\rangle_0 \equiv \mathrm{Tr}_E[\hat{E}_i\,\hat{\rho}_E(0)] = 0, \tag{1.10}$$

This is achieved by assuming that the total system starts in a **separable** product state of the form

$$\hat{\rho}_{T,I}(0) = \hat{\rho}_{S,I}(0) \otimes \hat{\rho}_E(0), \tag{1.11}$$

Some textbooks impose the condition Eq. (1.10) as a starting assumption. Here a step further is taken. When it is not initially satisfied, it can *always* be enforced by redefining the Hamiltonian through a shift

$$\hat{H}_T = \hat{H}'_S + \hat{H}_E + \alpha\hat{H}'_{\mathrm{int}},$$

with

$$\hat{H}'_{\mathrm{int}} = \sum_i \hat{S}_i \otimes \hat{E}'_i, \qquad \hat{E}'_i = \hat{E}_i - \langle\hat{E}_i\rangle_0, \qquad \langle\hat{E}_i\rangle_0 \equiv \mathrm{Tr}_E[\hat{E}_i\,\hat{\rho}_E(0)],$$

and a correspondingly shifted system Hamiltonian

$$\hat{H}'_S = \hat{H}_S + \alpha \sum_i \hat{S}_i \langle \hat{E}_i \rangle_0.$$

This "renormalization" merely redefines system energy levels and does not affect dissipative structure. Hence, without loss of generality the shift is assumed to be performed so that $\langle \hat{E}_i \rangle_0 = 0$. Explicitly, the first-order term in Eq. (1.10) vanishes:

$$\sum_i \mathrm{Tr}_E\big[\hat{S}_i \otimes \hat{E}_i, \hat{\rho}_{S,I}(0) \otimes \hat{\rho}_E(0)\big] = \sum_i \big(\hat{S}_i \hat{\rho}_{S,I}(0) - \hat{\rho}_{S,I}(0)\hat{S}_i\big)\, \mathrm{Tr}_E[\hat{E}_i\, \hat{\rho}_E(0)] = 0,$$

which implements that a static bath mean field can be absorbed into $\hat{H}_S$. The reduced equation of motion becomes

$$\begin{aligned}\frac{d}{dt}\hat{\rho}_{S,I}(t) &= -i\,\alpha\, \mathrm{Tr}_E\big[\, \hat{H}_{\mathrm{int},I}(t), \hat{\rho}_{T,I}(0)\,\big] - \alpha^2 \int_0^t ds\, \mathrm{Tr}_E\big[\, \hat{H}_{\mathrm{int},I}(t), [\hat{H}_{\mathrm{int},I}(s), \hat{\rho}_{T,I}(t)]\,\big] \\ &= -\alpha^2 \int_0^t ds\, \mathrm{Tr}_E\big[\, \hat{H}_{\mathrm{int},I}(t), [\hat{H}_{\mathrm{int},I}(s), \hat{\rho}_{T,I}(t)]\,\big]. \end{aligned} \tag{1.12}$$

To make the time-translation structure explicit, set $s' = t - s$ (equivalently, $ds' = -ds$). When s runs from 0 to t, s' runs from t to 0. Reversing the limits removes the minus sign, so the integration range remains $[0, t]$. Using this change and expanding the double commutator,

$$\big[A, [B, X]\big] = ABX - AXB - BXA + XBA,$$

Eq. (1.12) can be rewritten in the form

$$\begin{aligned}\frac{d}{dt}\hat{\rho}_{S,I}(t) = \alpha^2 \int_0^t ds\, \mathrm{Tr}_E\Big\{ &\hat{H}_{\mathrm{int},I}(t)\big[\hat{H}_{\mathrm{int},I}(t-s)\hat{\rho}_{T,I}(t) - \hat{\rho}_{T,I}(t)\, \hat{H}_{\mathrm{int},I}(t-s)\big] \\ &- \big[\hat{H}_{\mathrm{int},I}(t-s)\hat{\rho}_{T,I}(t) - \hat{\rho}_{T,I}(t)\, \hat{H}_{\mathrm{int},I}(t-s)\big]\hat{H}_{\mathrm{int},I}(t) \Big\}.\end{aligned}$$

Next, the assumption of Eq. (1.11) is strengthened by requiring that throughout the evolution the total state remains close to a factorized form $\hat{\rho}_{T,I}(t) \approx \hat{\rho}_{S,I}(t) \otimes \hat{\rho}_{E,I}(t)$. This can again be stated as the **Born approximation** of weak coupling. Collecting all system operators and all environment operators, and inserting the interaction Hamiltonian Eq. (1.6) explicitly, the following expression is obtained. Operators at time $t - s$ are tracked with index i' and at time t with i

$$\begin{aligned}\frac{d}{dt}\hat{\rho}_{S,I}(t) = \alpha^2 \sum_{i,i'} \int_0^t ds \Big\{ &\mathrm{Tr}_E\big[\hat{S}_{i,I}(t)\hat{S}_{i',I}(t-s)\hat{\rho}_{S,I}(t) \otimes \hat{E}_{i,I}(t)\hat{E}_{i',I}(t-s)\hat{\rho}_{E,I}(t)\big] - \\ &\mathrm{Tr}_E\big[\hat{S}_{i,I}(t)\hat{\rho}_{S,I}(t)\hat{S}_{i',I}(t-s) \otimes \hat{E}_{i,I}(t)\hat{\rho}_{E,I}(t)\hat{E}_{i',I}(t-s)\big] - \\ &\mathrm{Tr}_E\big[\hat{S}_{i',I}(t-s)\hat{\rho}_{S,I}(t)\hat{S}_{i,I}(t) \otimes \hat{E}_{i',I}(t-s)\hat{\rho}_{E,I}(t)\hat{E}_{i,I}(t)\big] + \\ &\mathrm{Tr}_E\big[\hat{\rho}_{S,I}(t)\hat{S}_{i',I}(t-s)\hat{S}_{i,I}(t) \otimes \hat{\rho}_{E,I}(t)\hat{E}_{i',I}(t-s)\hat{E}_{i,I}(t)\big] \Big\}.\end{aligned}$$

Since the trace only acts on the environment, the system operators can be taken out of the trace, and the two-point correlation functions are defined using Eq. (1.25)

$$C_{ii'}(t, s) \equiv \langle \hat{E}_{i,I}(t)\hat{E}_{i',I}(t-s)\rangle = \mathrm{Tr}_E\big[\hat{E}_{i,I}(t)\hat{E}_{i',I}(t-s)\, \hat{\rho}_{E,I}(t)\big]. \tag{1.13}$$

The term 'two-point' indicates that the bath operators $\hat{E}_i$ and $\hat{E}_{i'}$ are evaluated at two different times. Up to this point, the derivation has been general and no assumption of stationarity was required. The correlator can depend on both times through $\hat{\rho}_{E,I}(t)$. Now the bath is taken to be stationary, i.e. $[\hat{H}_E, \hat{\rho}_E(0)] = 0$. This implies $\hat{\rho}_{E,I}(t) = \hat{\rho}_E(0)$ and therefore

$$C_{ii'}(t,s) = C_{ii'}(t-s).$$

To make this dependence explicit we introduce the time difference

$$\tau \equiv t - s,$$

and subsequently write $C_{ii'}(\tau)$ in place of $C_{ii'}(t,s)$.

Units. With $\hat{H}_{\text{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i$ and $\hbar = 1$, $\hat{H}_{\text{int}}$ has units of time^{-1}, hence $[\hat{S}_i][\hat{E}_i] = \text{time}^{-1}$. If $\hat{S}_i$ is chosen dimensionless, then $[\hat{E}_i] = \text{time}^{-1}$ and therefore $[C_{ii'}] = \text{time}^{-2}$.

If the bath operators are Hermitian, i.e. $\hat{E}_i^\dagger = \hat{E}_i$, and the bath state is stationary (i.e., $[\hat{H}_E, \hat{\rho}_E(0)] = 0$), the cyclic property of the trace gives the conjugation relation

$$C^*_{ii'}(\tau) = \tilde{C}_{ii'}(\tau) \equiv \langle \hat{E}_{i',I}(t-\tau)\hat{E}_{i,I}(t)\rangle,$$

In the interaction picture, stationarity implies $\hat{\rho}_{E,I}(t) = \hat{\rho}_{E,I}(0) = \hat{\rho}_E(0)$ and the correlators depend only on the time difference $\tau = t - s$. With $\hat{E}_{i,I}(t) = e^{i\hat{H}_E t}\hat{E}_i e^{-i\hat{H}_E t}$ this implies the further symmetry condition

$$C^*_{ii'}(-\tau) = C_{ii'}(\tau).$$

and thus finally obtain the desired form of the Redfield equation

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = \alpha^2 \sum_{i,i'} \int_0^t ds \Big\{ C_{ii'}(t-s)\big[\hat{S}_{i,I}(t), \hat{S}_{i',I}(t-s)\hat{\rho}_{S,I}(t)\big] + \text{H.c.} \Big\}. \tag{1.14}$$

This form shows explicitly that the system evolution depends only on bath correlation functions evaluated along the free bath dynamics.

1.1.5 Markov Approximation

However Eq. (1.14) is still not Markovian, since it carries an explicit integration over time t. An additional approximation is therefore introduced. Besides weak coupling, assume that the bath correlation functions decay on a characteristic time scale τ_E that is much shorter than the relevant system evolution time τ_S, i.e. $\tau_E \ll \tau_S$. In particular, $C_{ii'}(\tau)$ becomes negligible for $\tau \gtrsim \tau_E$, so the integrand has support only on a short memory window. If the reduced state varies slowly on this window, one may approximate it as constant over the integration range and extend the upper limit to infinity, which yields a Markovian generator for $t \gg \tau_E$.

$$\int_0^t ds\, C_{ii'}(t-s) f(s) \;\longrightarrow\; \int_0^\infty d\tau\, C_{ii'}(\tau) f(t), \qquad (t \gg \tau_E),$$

where the slow variation of $\hat{\rho}_{S,I}(t)$ and of the system operators on the time scale τ_E has been used. Applying this rule to Eq. (1.12) yields a *time-local* Markovian Redfield generator. In the interaction picture the operators still carry explicit t-dependence, but the kernel is time-translation invariant and depends only on τ, so the generator has no explicit dependence on absolute time beyond the system operators themselves

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -\alpha^2 \sum_{i,i'} \int_0^\infty d\tau \Big(C_{ii'}(\tau)[\hat{S}_{i,I}(t), \hat{S}_{i',I}(t-\tau)\hat{\rho}_{S,I}(t)] + \text{H.c.} \Big). \tag{1.15}$$

Here $\hat{S}_{i,I}(t-\tau)$ denotes the free interaction-picture evolution under $\hat{H}_S$, so the explicit $(t-\tau)$ dependence reflects free system dynamics rather than any memory in the kernel. This step encapsulates the loss of memory: the generator now includes a very short time window τ_E where the system operators can be approximated as constant (frozen at time $\hat{\rho}_{S,I}(t)$). Equation Eq. (1.15) is the Redfield master equation in the interaction picture under the Born (weak-coupling) and Markov (short-memory) approximations. In this form, all environmental effects are encoded in the bath correlation functions $C_{ii'}(\tau)$, which are analyzed in the next section.

1.2 Environmental Correlators and Spectral Properties

In what follows, the origin and key properties of bath correlation functions (defined by Eq. (1.13)) are outlined, their spectral Fourier representation is introduced, and the simultaneous encoding of emission and absorption processes is discussed. In particular, the decay of $C_{ii'}(\tau)$ on the bath correlation time τ_E quantifies how rapidly the environment "forgets", and the Markov approximation used above is justified when τ_E is much shorter than the relevant system evolution time scales. The connection between $C_{ii'}(\tau)$ and the spectral density is direct and experimentally accessible. Modern noise spectroscopy techniques enable direct measurement of bath spectral properties from coherence-decay observations without requiring pulse sequences or prior knowledge of the system-bath coupling constants—providing parameter-free experimental access to the bath structure that drives decoherence in the e.g. molecular systems [17].

1.2.1 Fourier Representation and Secular Approximation

For a stationary thermal bath in equilibrium, where its state is given by the Gibbs state (see Sec. 1.3.1), the Kubo–Martin–Schwinger (KMS) condition relates correlations at positive and negative times via a shift into the complex time plane,

$$C_{ii'}(\tau) \equiv \langle \hat{E}_{i,I}(\tau)\hat{E}_{i',I}(0)\rangle = \langle \hat{E}_{i,I}(0)\hat{E}_{i',I}(\tau+i\beta)\rangle \equiv C_{i'i}(+\tau+i\beta). \tag{1.16}$$

A short proof of this relation in the eigenbasis of the bath Hamiltonian is as follows

$$\begin{aligned}
C_{ii'}(\tau+i\beta) &= \mathrm{Tr}(\hat{\rho}_\beta\, \hat{E}_i\, \hat{E}_{i',I}(\tau+i\beta)) \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_n}\, \langle n|\, \hat{E}_i\, |m\rangle\, \langle m|\, \hat{E}_{i',I}(\tau+i\beta)\, |n\rangle \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_n} e^{i(E_m-E_n)(\tau+i\beta)} (\hat{E}_i)_{nm}(\hat{E}_{i'})_{mn} \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_m} e^{i(E_n-E_m)\tau} (\hat{E}_{i'})_{nm}(\hat{E}_i)_{mn} \\
&= C_{i'i}(\tau).
\end{aligned}$$

Eigenoperator Decomposition

To proceed analytically and to connect with relaxation pathways, the system coupling operators are decomposed into eigenoperators of the system Hamiltonian. Let $\hat{H}_S = \sum_\epsilon \epsilon\, \hat{\Pi}_\epsilon$ be the spectral resolution with projectors $\hat{\Pi}_\epsilon$. The Bohr frequencies are defined as $\omega = \epsilon' - \epsilon$ and the corresponding interaction operators in the eigenbasis are

$$\hat{S}_i(\omega) = \sum_{\epsilon'-\epsilon=\omega} \hat{\Pi}_\epsilon\, \hat{S}_i\, \hat{\Pi}_{\epsilon'}.$$

In the interaction picture these acquire simple oscillatory phases

$$\hat{S}_{i,I}(t) = \sum_\omega e^{-i\omega t} \hat{S}_i(\omega), \qquad \hat{S}_{i',I}(t-\tau) = \sum_{\omega'} e^{-i\omega'(t-\tau)} \hat{S}_{i'}(\omega'). \tag{1.17}$$

Note that a time-independent system Hamiltonian $\hat{H}_S$ is assumed here. The derivation can in principle be extended to time-dependent $\hat{H}_S(t)$, for example in spectroscopy where external driving fields lead to an explicitly time-dependent system Hamiltonian [4]. Generalizations of Markovian master-equation derivations to driven systems are discussed in [18].

Inserting Eq. (1.17) into Eq. (1.15) produces sums over oscillatory factors $e^{-i(\omega-\omega')t}$ multiplying integrals of $C_{ii'}(\tau)e^{i\omega'\tau}$. The inner commutator in Eq. (1.15) becomes

$$[\hat{S}_{i,I}(t),\, \hat{S}_{i',I}(t-\tau)\, \hat{\rho}_{S,I}(t)] = \sum_{\omega,\omega'} e^{-i(\omega-\omega')t} e^{+i\omega'\tau} \left[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{S,I}(t)\right],$$

and the total Redfield equation reads

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -\alpha^2 \sum_{i,i'} \sum_{\omega,\omega'} e^{-i(\omega-\omega')t} \left(\int_0^\infty d\tau\, C_{ii'}(\tau)\, e^{+i\omega'\tau} \right) \left[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{S,I}(t)\right] + \text{H.c.}.$$

The correlator is now expressed in the frequency domain and the noise-power spectrum of the environment is defined [19]. To avoid confusion with the system coupling operators $\hat{S}_i$, we denote the bath spectrum by $\mathcal{S}(\omega)$ (common in the literature as $S(\omega)$).

$$\mathcal{S}(\omega) \equiv \int_{-\infty}^{\infty} d\tau\, C(\tau)\, e^{+i\omega\tau}, \tag{1.18}$$

Units. Since $[C] = \text{time}^{-2}$ for dimensionless $\hat{S}_i$ and $\hbar = 1$, the Fourier transform implies $[\mathcal{S}(\omega)] = [C]\,\text{time} = \text{time}^{-1}$.

This quantity tells how strongly bath fluctuations at frequency ω contribute to noise. For a stationary bath correlation function $C(\tau)$, the noise spectrum $\mathcal{S}(\omega)$ is by the Wiener–Khinchin relation the Fourier transform of $C(\tau)$ and real and non-negative $\mathcal{S}(\omega) \geq 0$ [20].

The KMS condition Eq. (1.16) can now be translated to frequency domain, where it imposes the detailed-balance symmetry

$$\mathcal{S}(-\omega) = e^{-\omega\beta}\mathcal{S}(\omega), \tag{1.19}$$

which ensures that upward and downward transition rates obey Boltzmann ratios.

The exact one-sided Fourier (Laplace-) transform of the bath correlation function can be split into the power spectrum and an energy shift $\lambda(\omega)$ as

$$\begin{aligned} \chi_{ii'}(\omega) &\equiv \int_0^\infty d\tau\, C_{ii'}(\tau)\, e^{+i\omega\tau} \\ &= \tfrac{1}{2}\mathcal{S}_{ii'}(\omega) + i\,\lambda_{ii'}(\omega), \end{aligned}$$

with

$$\mathcal{S}_{ii'}(\omega) = \chi_{ii'}(\omega) + \chi^*_{i'i}(\omega).$$

which reduce Sec. 1.2.1 to the compact form

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -\alpha^2 \sum_{i,i'} \sum_{\omega,\omega'} e^{-i(\omega-\omega')t}\, \chi_{ii'}(\omega') \left[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{S,I}(t)\right] + \text{H.c.}.$$

Going back to the Schrödinger picture, this yields

$$\frac{d}{dt}\hat{\rho}_S(t) = -i[\hat{H}_S + \hat{H}_{\text{LS}}, \hat{\rho}_S(t)] + \mathcal{D}_{\text{R}}[\hat{\rho}_S(t)],$$

with the Lamb-shift Hamiltonian

$$\hat{H}_{\text{LS}} = \sum_{\omega} \sum_{i,i'} \lambda_{ii'}(\omega)\, \hat{S}_i^\dagger(\omega)\hat{S}_{i'}(\omega),$$

which will be absorbed into a renormalized system Hamiltonian and disregarded in the subsequent derivation. The Redfield dissipator is

$$\mathcal{D}_{\mathrm{R}}[\hat{\rho}] = \frac{1}{2}\sum_{\omega,\omega'}\sum_{i,i'} e^{-i(\omega-\omega')t}\mathcal{S}_{ii'}(\omega')\left(\hat{S}_{i'}(\omega')\,\hat{\rho}\,\hat{S}_i^{\dagger}(\omega) - \hat{S}_i^{\dagger}(\omega)\hat{S}_{i'}(\omega')\,\hat{\rho}\right) + \text{H.c.}. \tag{1.20}$$

Matrix Elements and Redfield Tensor

In the eigenbasis of the Hamiltonian $\{|a\rangle\}$, taking $\langle a|\cdot|b\rangle$ of the master equation and grouping terms gives

$$\frac{d}{dt}\rho_{ab}(t) = -i\,\omega_{ab}\rho_{ab}(t) + \sum_{c,d} R_{abcd}\,\rho_{cd}(t),$$

where the Bloch–Redfield tensor, also implemented in QuTiP, is defined as

$$\begin{aligned} R_{abcd} = -\frac{1}{2}\sum_{i,i'}\Big\{&\delta_{bd}\sum_n S^i_{an}S^{i'}_{nc}\,\mathcal{S}_{ii'}(\omega_{cn}) - S^{i'}_{ac}S^i_{db}\,\mathcal{S}_{ii'}(\omega_{ca}) \\ &+ \delta_{ac}\sum_n S^i_{dn}S^{i'}_{nb}\,\mathcal{S}_{ii'}(\omega_{dn}) - S^{i'}_{ac}S^i_{db}\,\mathcal{S}_{ii'}(\omega_{db})\Big\}. \end{aligned} \tag{1.21}$$

with S^i the system coupling operators and $\mathcal{S}_{ii'}(\omega)$ the bath spectra[19].

Secular Approximation

The oscillatory prefactors $e^{-i(\omega-\omega')t}$ in Eq. (1.20) average to zero on coarse-grained times Δt satisfying $\tau_E \ll \Delta t \ll 1/|\omega-\omega'|$ whenever $\omega \neq \omega'$. Keeping only terms with $\omega_{ab} = \omega_{cd}$ in Eq. (1.21) removes such fast-rotating couplings and ensures a block-diagonal structure in the Redfield tensor R_{abcd}.

This yields the *secular* Redfield equation in Lindblad form

$$\begin{aligned} \frac{d}{dt}\hat{\rho}_S(t) &= -i[\hat{H}_S + \hat{H}_{\mathrm{LS}}, \hat{\rho}_S(t)] \\ &\quad + \sum_{i,i'}\sum_{\omega}\gamma_{ii'}(\omega)\left(\hat{S}_{i'}(\omega)\hat{\rho}_S(t)\hat{S}_i^{\dagger}(\omega) - \tfrac{1}{2}\{\hat{S}_i^{\dagger}(\omega)\hat{S}_{i'}(\omega), \hat{\rho}_S(t)\}\right), \end{aligned}$$

where $\gamma_{ii'}(\omega) = 2\mathrm{Re}\,\chi_{ii'}(\omega)$. This form is guaranteed to preserve complete positivity provided the matrix $[\gamma_{ii'}(\omega)]_{i,i'}$ is positive semidefinite for each frequency ω. Without the secular approximation, Eq. (1.20) need not generate a completely positive dynamical map, explaining the caution required when applying the full Redfield equation.

For a thermal bath in equilibrium, the bath correlation functions satisfy the KMS condition. In the weak-coupling limit with a full secular approximation, the resulting Lindblad generator has the Gibbs state as a stationary state (cf. Equation (1.24)) [21, 22]

$$\mathcal{L}_{\mathrm{sec}}[\hat{\rho}_{\mathrm{Gibbs}}] = 0.$$

By contrast, the non-secular (Bloch-)Redfield generator retains terms with $\omega \neq \omega'$, which couple populations and coherences in the energy basis. Without full secularization, quantum detailed balance and Gibbs stationarity are not guaranteed, even for a single thermal bath satisfying KMS [1, 23]. In weak coupling, the stationary state typically differs from $\hat{\rho}_{\mathrm{Gibbs}}$ by corrections of second order in the system–bath coupling (and may exhibit bath-induced population shifts and coherences).

An important exception is pure dephasing, where the coupling operator commutes with the system Hamiltonian, $[\hat{A}, \hat{H}_S] = 0$. In that case there is no energy exchange and $\hat{\rho}_\beta$ remains stationary also without secularization [1].

Uncorrelated Hermitian Couplings

If the baths are uncorrelated and $S_i = S_i^\dagger$:

$$\mathcal{S}_{ii'}(\omega) = \delta_{ii'}\mathcal{S}_i(\omega) \implies$$
$$R_{abcd} = -\frac{1}{2}\sum_i \Big\{ \delta_{bd}\sum_n S^i_{an}S^i_{nc}\mathcal{S}_i(\omega_{cn}) - S^i_{ac}S^i_{db}\mathcal{S}_i(\omega_{ca}) + \delta_{ac}\sum_n S^i_{dn}S^i_{nb}\mathcal{S}_i(\omega_{dn}) - S^i_{ac}S^i_{db}\mathcal{S}_i(\omega_{db}) \Big\}.$$

1.2.2 Physical Interpretation: Emission and Absorption

Emission and absorption are both automatically included because $\mathcal{S}_{ii'}(\omega)$ contains positive- and negative-frequency components related by Eq. (1.19). Positive frequencies ($\omega > 0$) describe system energy loss (emission), while negative frequencies ($\omega < 0$) describe system energy gain (absorption). Detailed balance follows immediately: the ratio of absorption to emission contributions at frequency $\omega > 0$ is $e^{-\omega\beta}$. In the high-temperature limit ($\beta \to 0$), this simplifies to $\mathcal{S}(-\omega) \approx \mathcal{S}(\omega)$, making the power spectrum symmetric around $\omega = 0$ and rendering emission and absorption rates nearly equal.

The explicit relation between $J(\omega)$, the correlator, and the full spectrum is derived in the next section (see Eq. (1.33)).

1.3 Harmonic Oscillator Baths and Explicit Correlation Functions

The previous section established that the reduced dynamics is governed by the bath correlation functions $C_{ij}(\tau)$. To make the Redfield equation practical, these correlators must be evaluated for a concrete microscopic model. Here the focus is on the harmonic-oscillator bath, a widely used model that captures electromagnetic field fluctuations, phonons, and other bosonic environments.

The task now is to calculate the bath correlation function $C(\tau)$ for a concrete model. Recall that $C(\tau) = \langle E(t+\tau)E(t)\rangle$ measures how bath fluctuations at two different times are correlated in thermal equilibrium. For the harmonic oscillator bath, these correlators can be computed exactly, revealing how $C(\tau)$ connects to experimentally relevant spectral densities and governs the system's approach to thermal equilibrium.

For these explicit calculations, the following mathematical result is useful.

Geometric Series

The infinite geometric series

$$\mathcal{G} = a + ar + ar^2 + ar^3 + \cdots = \sum_{n=0}^{\infty} ar^n,$$

converges to

$$\mathcal{G} = \frac{a}{1-r} \quad \text{for } |r| < 1. \tag{1.22}$$

Differentiating Eq. (1.22) with respect to r gives

$$\sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}, \quad |r| < 1. \tag{1.23}$$

The formal correlation functions introduced above are now specialized to the paradigmatic bosonic bath. This leads naturally to spectral-density representations and the standard phenomenological models such as Ohmic, sub-/super-Ohmic, and Drude–Lorentz.

1.3.1 Bosonic Environment and Gibbs State

A bosonic bath is modeled as a (large) collection of independent harmonic oscillators in thermal equilibrium:

The thermal state of the bosonic bath is given by the Gibbs density matrix

$$\hat{\rho}_{\text{Gibbs}} = \frac{e^{-\beta \hat{H}_E}}{\text{Tr}[e^{-\beta \hat{H}_E}]}, \tag{1.24}$$

so that for any bath operator $\hat{A}$,

$$\langle \hat{A} \rangle = \text{Tr}(\hat{\rho}_{\text{Gibbs}} \hat{A}) = \frac{1}{Z} \sum_n e^{-\beta E_n} A_{nn}, \tag{1.25}$$

in the energy eigenbasis of $\hat{H}_E$, where $Z = \text{Tr}[e^{-\beta \hat{H}_E}]$ and $\beta = 1/(k_B T)$ is the inverse temperature of the environment. The bath Hamiltonian consists of independent harmonic oscillators

$$\hat{H}_E = \sum_k \omega_k \left(\hat{b}_k^\dagger \hat{b}_k + \tfrac{1}{2} \right), \tag{1.26}$$

and the average energy of mode k is

$$E_k = \omega_k (n_k + \tfrac{1}{2}),$$

with $n_k = \langle \hat{b}_k^\dagger \hat{b}_k \rangle$ the Bose–Einstein occupation number.

The zero-point contribution $E_{\text{vac}} = \frac{1}{2} \sum_k \omega_k$ adds only a mode-independent constant to the bath energy. Constant energy offsets commute with all observables, so they neither modify thermal populations nor enter the correlation functions that drive the system dynamics. This term is therefore dropped in the dynamical equations (standard for normal ordering, where raising operators are placed to the left of lowering operators). For completeness, the $\frac{1}{2}$ is retained in the partition function Z_k for normalization, but it cancels out of all correlation functions and rates. In other words, only energy *differences* influence the evolution.

Single Mode Partition Function and Occupation

For a single mode k, the thermal state reads

$$\hat{\rho} = \frac{e^{-\beta \omega_k \hat{b}_k^\dagger \hat{b}_k}}{Z_k},$$

with partition function

$$\begin{aligned} Z_k \equiv \text{Tr}\left[e^{-\beta \hat{H}_E} \right] &= \sum_{m=0}^{\infty} \langle m | \, e^{-\beta \omega_k (\hat{b}_k^\dagger \hat{b}_k + \frac{1}{2})} \, | m \rangle \\ &= \frac{e^{-\beta \omega_k / 2}}{1 - e^{-\beta \omega_k}}. \end{aligned}$$

The Bose–Einstein average occupation number follows using Eq. (1.23) and is given by

$$\begin{aligned} n_k = \langle \hat{b}_k^\dagger \hat{b}_k \rangle_{\text{th}} &= \text{Tr}\left[\hat{b}_k^\dagger \hat{b}_k \frac{e^{-\beta \hat{H}_E}}{Z_k}\right] \\ &= \frac{\text{Tr}\left[\hat{b}_k^\dagger \hat{b}_k e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k}\right]}{Z_k} \\ &= \frac{\sum_{m=0}^{\infty} \langle m| \hat{b}_k^\dagger \hat{b}_k e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k} |m\rangle}{\frac{e^{-\beta\omega_k/2}}{1-e^{-\beta\omega_k}}} \\ &= \frac{e^{-\beta\omega_k}}{1-e^{-\beta\omega_k}} \\ &= \frac{1}{e^{\beta\omega_k}-1}. \end{aligned} \tag{1.27}$$

Since the bath is composed of many independent modes, this calculation can be done for every mode, and the total partition function factorizes to

$$Z_{\text{bath}} = \prod_k Z_k = \prod_k \frac{e^{-\beta\omega_k/2}}{1-e^{-\beta\omega_k}}.$$

With the thermal properties of the bath established, attention can be turned to its coupling to a small system of interest. The bath correlation functions from Eq. (1.13) are therefore calculated for the simple case of a single coupling operator $\hat{E}$. The derivation follows and slightly expands [24].

1.3.2 Microscopic Form of the Bath Correlator

With linear system–bath coupling to oscillator displacements, the Caldeira–Leggett model [1, 25] gives the single bath coupling operator as

$$\hat{E} = \sum_{n=1}^{\infty} c_n \hat{x}_n, \qquad \hat{x}_n = \sqrt{\frac{1}{2m_n\omega_n}}(\hat{b}_n + \hat{b}_n^\dagger).$$

In the interaction picture the bath operators evolve as $\hat{E}_I(t) = e^{i\hat{H}_E t}\hat{E}e^{-i\hat{H}_E t}$ with $\hat{H}_E$ from Eq. (1.26), yielding

$$\hat{E}_I(0) = \sum_n c_n \sqrt{\frac{1}{2m_n\omega_n}}(\hat{b}_n + \hat{b}_n^\dagger), \tag{1.28}$$

$$\hat{E}_I(\tau) = \sum_n c_n \sqrt{\frac{1}{2m_n\omega_n}}\Big(\hat{b}_n e^{-i\omega_n\tau} + \hat{b}_n^\dagger e^{i\omega_n\tau}\Big). \tag{1.29}$$

The thermal correlation function for this single relevant bath operator is

$$C(\tau) = \langle \hat{E}_I(\tau)\hat{E}_I(0)\rangle.$$

Inserting Eqs. (1.28) and (1.29) yields a double sum over bath modes,

$$\begin{aligned} C(\tau) &= \sum_{n,m} \frac{c_n c_m}{\sqrt{4m_n m_m \omega_n \omega_m}} \Big\langle \big(\hat{b}_n e^{-i\omega_n\tau} + \hat{b}_n^\dagger e^{i\omega_n\tau}\big)(\hat{b}_m + \hat{b}_m^\dagger)\Big\rangle, \\ &= \sum_{n,m} \frac{c_n c_m}{\sqrt{4m_n m_m \omega_n \omega_m}} \Big(e^{-i\omega_n\tau}\langle \hat{b}_n \hat{b}_m^\dagger\rangle + e^{i\omega_n\tau}\langle \hat{b}_n^\dagger \hat{b}_m\rangle\Big), \end{aligned}$$

where mixed terms such as $\langle \hat{b}_n \hat{b}_m \rangle$ vanish for a thermal state of uncoupled oscillators. Doing similar calculations to Eq. (1.27) gives

$$\langle \hat{b}_n \hat{b}_m^\dagger \rangle = \delta_{nm}(n_n + 1) \quad \text{and} \quad \langle \hat{b}_n^\dagger \hat{b}_m \rangle = \delta_{nm} n_n,$$

with $n_n = \langle \hat{b}_n^\dagger \hat{b}_n \rangle = 1/(e^{\beta\omega_n} - 1)$ given by Eq. (1.27). Using these results, the discrete form of the bath correlation function is obtained as

$$C(\tau) = \sum_n \frac{c_n^2}{2m_n\omega_n} \Big[(n_n + 1)e^{-i\omega_n\tau} + n_n e^{i\omega_n\tau} \Big].$$

1.3.3 Spectral Density Representation

To connect this discrete expression with the continuum form, note that the correlator can be expanded by writing $e^{\pm i\omega_n\tau} = \cos(\omega_n\tau) \pm i\sin(\omega_n\tau)$. This yields

$$\begin{aligned} C(\tau) &= \sum_n \frac{c_n^2}{2m_n\omega_n} \Big[(n_n + 1)(\cos(\omega_n\tau) - i\sin(\omega_n\tau)) + n_n(\cos(\omega_n\tau) + i\sin(\omega_n\tau)) \Big] \\ &= \sum_n \frac{c_n^2}{2m_n\omega_n} \Big[(2n_n + 1)\cos(\omega_n\tau) - i\sin(\omega_n\tau) \Big]. \end{aligned}$$

Since $2n_n + 1 = 2/(e^{\beta\omega_n} - 1) + 1 = \coth(\beta\omega_n/2)$ (recall $\coth(x/2) = (e^x + 1)/(e^x - 1)$), this can be rewritten as

$$C(\tau) = \sum_n \frac{c_n^2}{2m_n\omega_n} \Big[\coth\left(\frac{\beta\omega_n}{2}\right) \cos(\omega_n\tau) - i\sin(\omega_n\tau) \Big]. \tag{1.30}$$

To move from discrete modes to a continuum description, introduce the spectral density, which encodes both the mode density and coupling strength,

$$J(\omega) = \pi \sum_n \frac{c_n^2}{2m_n\omega_n} \delta(\omega - \omega_n),$$

which summarizes how strongly each bath frequency couples to the system. Here it is assumed that the bath has discrete modes ω_n and each couples to the system with strength $\propto c_n$. By construction, the delta-function representation implies the distributional identity that for any smooth test function $f(\omega)$,

$$\sum_n \frac{c_n^2}{2m_n\omega_n} f(\omega_n) = \frac{1}{\pi} \int_0^\infty d\omega \, J(\omega) f(\omega). \tag{1.31}$$

This simply means that the discrete mode sum can be replaced by a continuum integral once $J(\omega)$ is defined. We then choose the specific function $f(\omega) = \coth(\beta\omega/2)\cos(\omega\tau) - i\sin(\omega\tau)$ because it is exactly the factor appearing in the discrete correlator. Applying the identity with this f yields a representation of the bath correlator in a continuum limit

$$C(\tau) = \int_0^\infty d\omega \, \frac{J(\omega)}{\pi} \Big[\coth\left(\frac{\beta\omega}{2}\right) \cos(\omega\tau) - i\sin(\omega\tau) \Big]. \tag{1.32}$$

The cosine term is even in τ and captures symmetrized noise, while the sine term is odd and encodes dissipation. In the macroscopic limit, the mode set becomes dense, and the spectral density $J(\omega)$ can be expressed as $J(\omega) = \rho(\omega)g(\omega)^2$, where $\rho(\omega)$ is the density of states and $g(\omega)$ the form factor. The spectral density thus fully characterizes the environmental coupling: it determines which bath frequencies exist and how strongly each frequency couples to the system. The real part of $C(\tau)$ describes symmetrized noise, while the imaginary part describes dissipation [1, 2, 26].

The parametric choice of $J(\omega)$, i.e., the bath spectral density, fixes whether noise is dominated by low-frequency components and whether the bath is smooth or has sharp features [1, 2]. This will be illustrated visually in Sec. 1.4, see in particular Fig. 1.2 and Fig. 1.4.

The bath correlation function $C(\tau)$ is the key quantity determining the system dynamics in the Redfield equation. It is fully specified by the spectral density $J(\omega)$ and the temperature-dependent factor $\coth(\beta\omega/2)$, which weights each frequency according to thermal occupation. The spectral density is also directly related to the noise power spectrum (cf. Equation (1.18)):

$$\mathcal{S}(\omega) = \int_{-\infty}^{\infty} d\tau\, C(\tau) e^{+i\omega\tau} = 2\pi J(|\omega|) \begin{cases} n_{\text{th}}(\omega) + 1, & \omega > 0, \\ n_{\text{th}}(|\omega|), & \omega < 0, \end{cases} \tag{1.33}$$

Here $J(\omega)$ is defined for $\omega > 0$, and the thermal occupation is $n_{\text{th}}(\omega) = 1/(e^{\beta\omega} - 1)$. The piecewise form makes explicit that $\omega > 0$ corresponds to emission and $\omega < 0$ to absorption, with the detailed-balance ratio fixed by $e^{-\beta\omega}$. Thus, specifying either the correlator $C(\tau)$, the noise power spectrum $\mathcal{S}(\omega)$, or the spectral density $J(\omega)$ together with the temperature β suffices to fully characterize the environmental coupling. The spectral density is often the most convenient choice, as it is a real-valued function of a single variable and can be directly related to microscopic models.

In the following, some of the most common choices for $J(\omega)$ are introduced.

1.4 Common choices of Spectral Densities

The main qualitative differences between bath models are set by the low-frequency scaling of $J(\omega)$ and by how high-frequency modes are regularized (cutoff). Both directly impact correlation times and the Markov regime.

1.4.1 Ohmic and Related Spectral Densities

The Redfield equation can also be connected to classical phenomenological models of dissipation. For an Ohmic spectral density, the low-frequency behavior is linear, and the spectral density is given by [2]

$$J(\omega) \propto \gamma\omega,$$

where γ is the damping constant. This corresponds to a classical friction-like dissipation. To ensure physical behavior, a cutoff is introduced

$$J(\omega) = \alpha \frac{\omega^s}{\omega_c^{s-1}} e^{-\omega/\omega_c}, \tag{1.34}$$

which enforces convergence of integrals such as Eq. (1.33). Here α is a dimensionless coupling constant, ω_c is the cutoff frequency, and s controls the low-frequency behavior of the bath, thereby effecting the timescale of the environment. The parameter s classifies the spectral density into three regimes: Ohmic ($s = 1$), *sub-Ohmic* ($s < 1$), and *super-Ohmic* ($s > 1$) [2, 19]. Sub-Ohmic baths exhibit slower fluctuations and longer memory, while super-Ohmic baths suppress low-frequency noise and emphasize fast, high-frequency modes. To make this separation explicit in the numerical illustrations below, it is convenient to measure all frequencies in units of a characteristic system scale ω_0 and to express bath parameters as dimensionless ratios (ω_c/ω_0, T/ω_0, α/ω_0). As shown in Fig. 1.2, the sub-Ohmic correlation function remains nonzero for longer times than the Ohmic and super-Ohmic cases.

For an Ohmic bath, increasing ω_c pushes bath weight to shorter times and reduces the bath memory time. This is precisely the physical content behind the Markov approximation and is illustrated in Fig. 1.3.

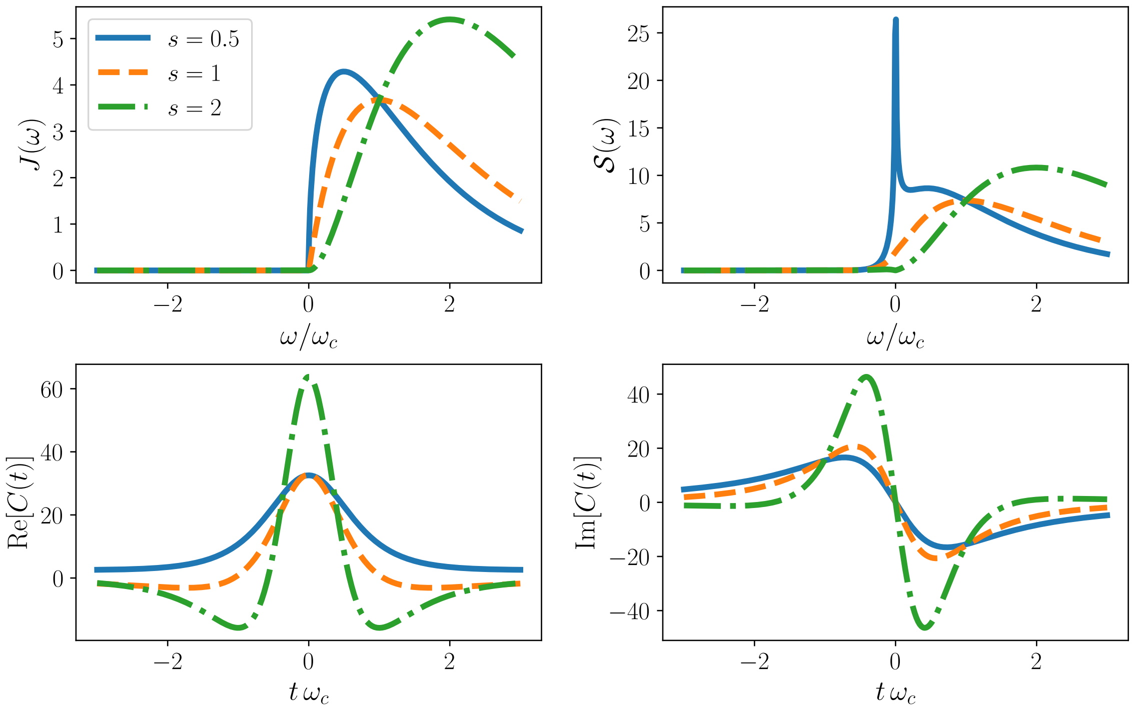

Figure 1.2: Comparison of sub-Ohmic ($s = 0.5$), Ohmic ($s = 1$), and super-Ohmic ($s = 2$) spectral densities, power spectra, and correlation functions. Parameters: $\omega_c/\omega_0 = 10$, $T/\omega_0 = 1$, $\alpha/\omega_0 = 1$.

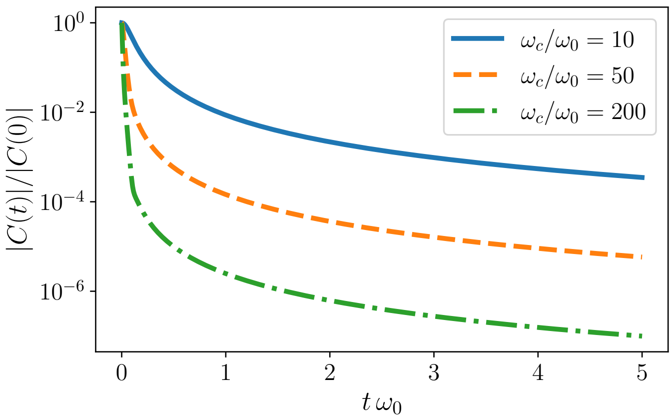

Figure 1.3: Normalized bath correlation magnitude $|C(t)|/|C(0)|$ for an Ohmic bath ($s = 1$), showing faster decay for larger cutoff ω_c/ω_0. Parameters: $T/\omega_0 = 10$, $\alpha/\omega_0 = 10^{-3}$.

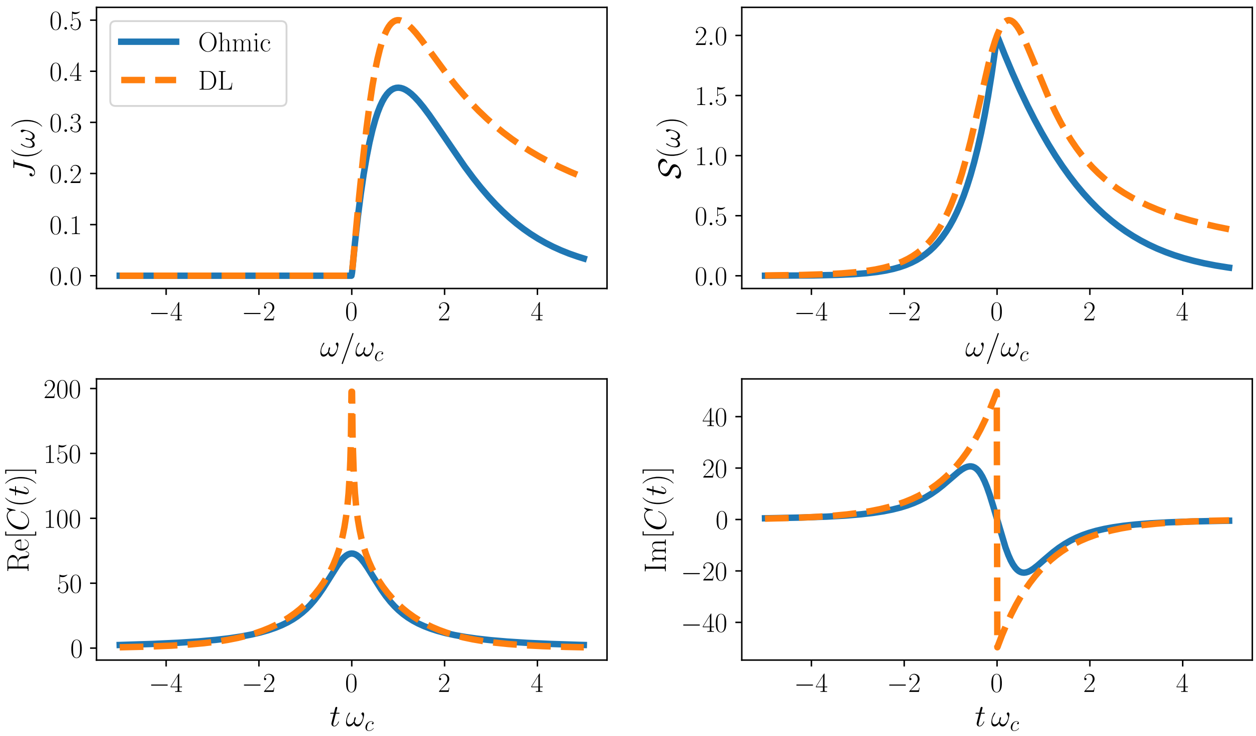

Figure 1.4: Comparison of representative bath models (Ohmic and Drude–Lorentz) showing spectral densities, associated power spectra, and time-domain correlation functions for coupling strength $\alpha = 0.1$, cutoff $\omega_c = 100$, and temperature $T = 100$. To make both models as comparable as possible, the Drude–Lorentz parameters are chosen consistently with the Ohmic-type convention via $\gamma = \omega_c$ and $\lambda = \alpha\,\omega_c/2$.

Another widely used model with linear low-frequency behavior is the Drude–Lorentz spectral density, which introduces a Lorentzian cutoff

$$J(\omega) = 2\lambda \frac{\omega\gamma}{\omega^2 + \gamma^2},$$

where λ is the reorganization energy, and γ is interpreted as a cutoff. This model represents an overdamped Brownian oscillator. A visual comparison between the Ohmic and Drude–Lorentz spectral densities, together with their corresponding correlation functions and power spectra, is shown in Fig. 1.4. The Drude–Lorentz model exhibits a longer high-frequency tail than the exponentially cut off Ohmic case.

Interestingly, when the electronic transition frequency substantially exceeds the characteristic bath frequencies, the *high-frequency tails* of $J(\omega)$ become dominant for determining population relaxation kinetics and must be carefully specified, even if spectral densities with different functional forms appear nearly identical in their peak structures [27].

1.4.2 Structured Spectral Densities

Beyond these overdamped models, underdamped Brownian-oscillator spectral densities are often used to represent discrete vibrational structure, or microscopic chain models such as the Rubin model, whose continuum limits recover Ohmic/Drude-type behavior [24, 28]. In practice, structured environments are frequently approximated by a sum of such components, such as Drude plus Brownian terms, which is also a standard starting point for hierarchical-equations-of-motion (HEOM) approaches [29, 30].

A bath is *structured* when $J(\omega)$ exhibits pronounced features near such transitions, such as sharp resonances, band edges, or steep slopes [1]. By contrast to the Ohmic cases, a resonant Lorentzian peak centered at a finite frequency ω_0 may be written as

$$J(\omega) = \lambda \frac{\gamma}{(\omega - \omega_0)^2 + \gamma^2}, \tag{1.35}$$

where γ controls the linewidth and λ the coupling strength. Such a spectrum concentrates weight near ω_0 and corresponds physically to a damped mode (e.g. a molecular vibration or cavity-like resonance) coupled to a broader background bath. The associated bath correlation function contains oscillatory contributions of the form $e^{-\gamma t}\cos(\omega_0 t)$, reflecting the finite-frequency structure. The distinction between structured and unstructured baths is independent of Markovianity.

In summary, environmental effects enter the Redfield framework through $J(\omega)$, which determines the bath correlation function $C(t) = \langle E(t)E(0)\rangle$ and thus the memory time τ_E (Eqs. (1.32) to (1.33)). Broad, slowly varying spectra typically support a Markovian description, whereas sharply structured features near system transitions can generate oscillatory and long-lived correlations, requiring more careful modeling beyond simple Markovian Redfield theory.

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